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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 25th July Morning Shift

JEE Main Question Paper with Answer Key

25 Jul 2022	90	22	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	20 questions
Mathematics	20 questions
Physics	28 questions
Out of syllabus	22 questions
Answer key	at the end

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Chemistry 20 questions

Q1. MCQ (Single Correct Answer)

SO_2Cl_2 on reaction with excess of water results into acidic mixture

$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$

16 moles of $NaOH$ is required for the complete neutralisation of the resultant acidic mixture. The number of moles of SO_2Cl_2 used is :

Open solution

(A) 16 (B) 8 (C) 4 (D) 2

Q2. MCQ (Single Correct Answer)

Which of the following sets of quantum numbers is not allowed?

Open solution

(A) $n = 3, 1 = 2, m_l = 0, s = +\frac{1}{2}$

(B) $n = 3, l = 2, m_l = -2, s = +\frac{1}{2}$

(C) $n = 3, l = 3, m_l = -3, s = -\frac{1}{2}$

(D) $n = 3, l = 0, m_l = 0, s = -\frac{1}{2}$

Q3. MCQ (Single Correct Answer)

The depression in freezing point observed for a formic acid solution of concentration $0.5\ mL\ L^{-1}$ is $0.0405^{\circ}C$. Density of formic acid is $1.05\ g\ mL^{-1}$. The Van't Hoff factor of the formic acid solution is nearly : (Given for water $k_f = 1.86\ k\ kg\ mol^{-1}$)

Open solution

(A) 0.8 (B) 1.1 (C) 1.9 (D) 2.4

Q4. MCQ (Single Correct Answer)

$20\ mL$ of $0.1\ M\ NH_4OH$ is mixed with $40\ mL$ of $0.05 MHCl$. The pH of the mixture is nearest to :

(Given : $K_b\ (NH_4OH) = 1 \times 10^{-5}, \log 2 = 0.30, \log 3 = 0.48, \log 5 = 0.69, \log 7 = 0.84, \log 11 = 1.04$)

Open solution

(A) 3.2 (B) 4.2 (C) 5.2 (D) 6.2

Q5. MCQ (Single Correct Answer)

The IUPAC nomenclature of an element with electronic configuration [Rn] $5f^{14}6d^{1}7s^{2}$ is :

Open solution

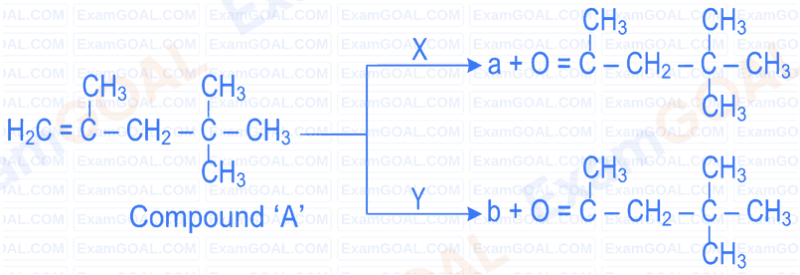
(A) Unnilbium (B) Unnilunium (C) Unnilquadium (D) Unniltrium

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Q6. MCQ (Single Correct Answer)

Open solution

A compound 'A' on reaction with 'X' and 'Y' produces the same major product but different by-products 'a' and 'b'. Oxidation of 'a' gives a substance produced by ants.



'X' and 'Y' respectively are

(A) $KMnO_4/H^+$ and dil. $KMnO_4$, 273 K

(B) $KMnO_4$ (dilute), 273 K and $KMnO_4/H^+$

(C) $KMnO_4/H^+$ and $O_3, H_2O/Zn$

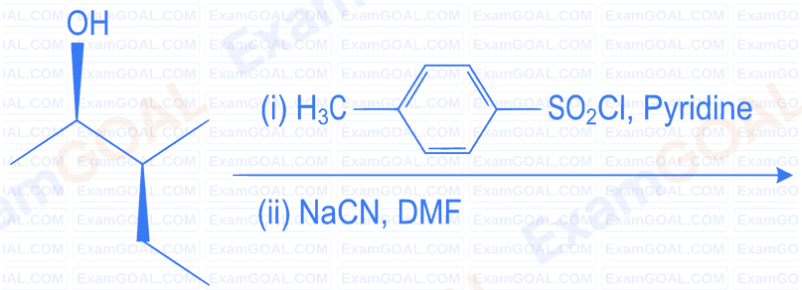
(D) $O_3, H_2O/Zn$ and $KMnO_4/H^+$

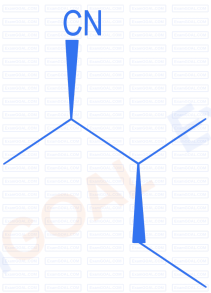
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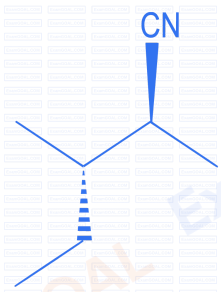
Q7. MCQ (Single Correct Answer)

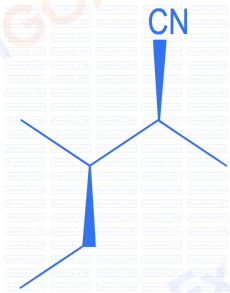
Open solution

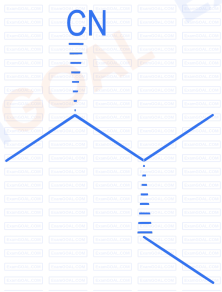
Most stable product of the following reaction is:



(A) 

(B) 

(C) 

(D) 

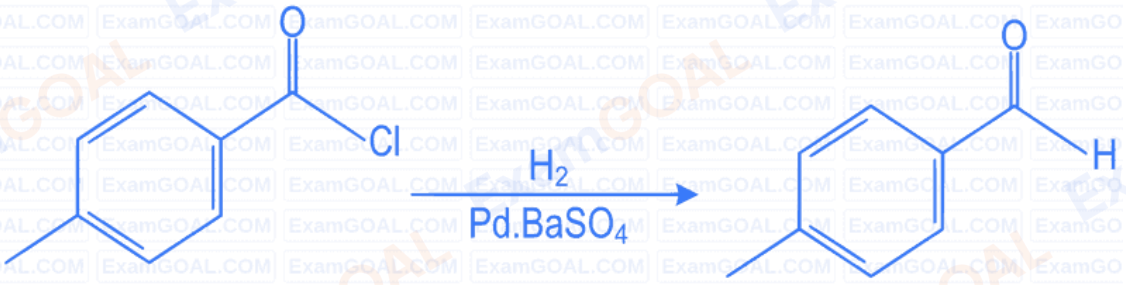
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Q8. MCQ (Single Correct Answer)

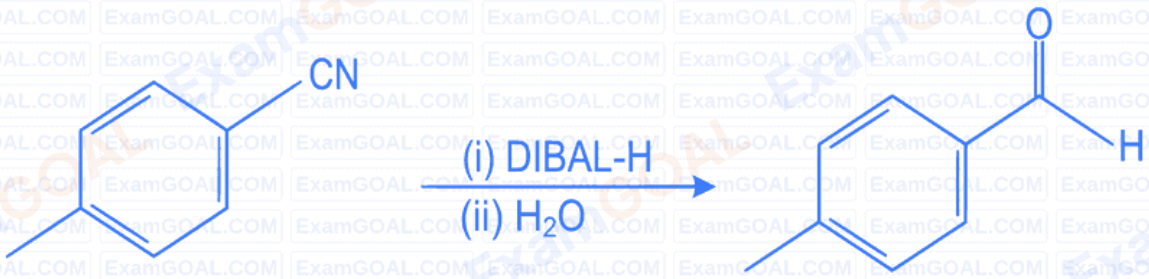
Open solution

Which one of the following reactions does not represent correct combination of substrate and product under the given conditions?

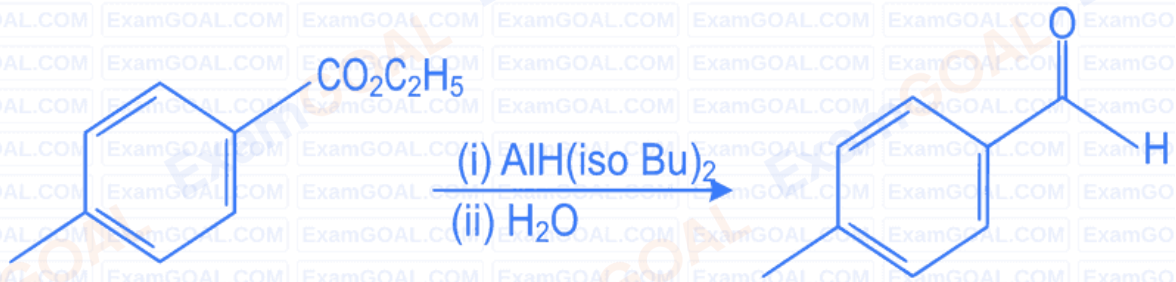
(A)



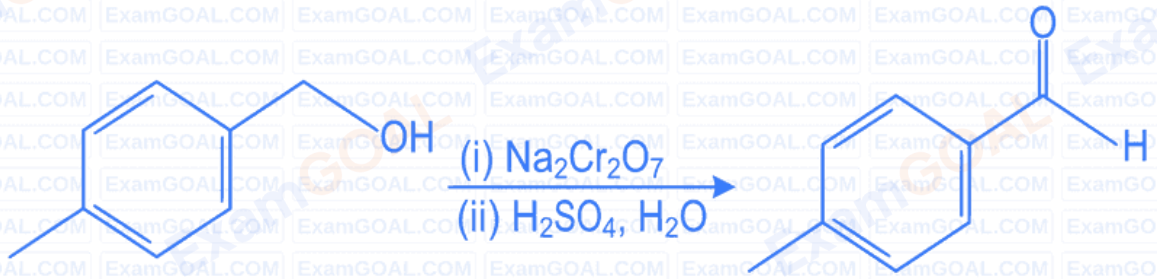
(B)



(C)



(D)



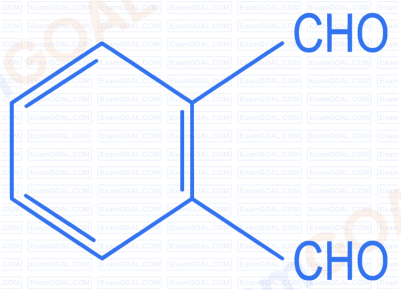
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Q9. MCQ (Single Correct Answer)

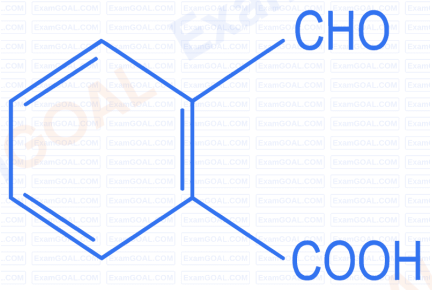
Open solution

An organic compound 'A' on reaction with NH_3 followed by heating gives compound B. Which on further strong heating gives compound C ($C_8H_5NO_2$). Compound C on sequential reaction with ethanolic KOH, alkyl chloride and hydrolysis with alkali gives a primary amine. The compound A is :

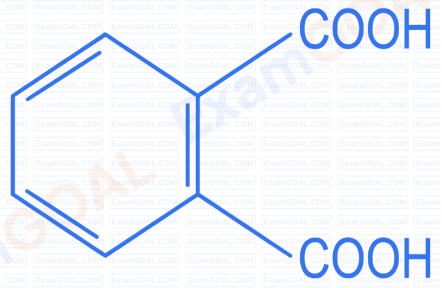
(A)



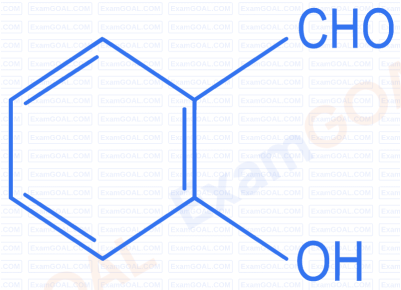
(B)



(C)



(D)



Q10. MCQ (Single Correct Answer)

Open solution

During the denaturation of proteins, which of these structures will remain intact?

(A) Primary (B) Secondary (C) Tertiary (D) Quaternary

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Q11. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : On heating with $KHSO_4$, glycerol is dehydrated and acrolein is formed.

Statement II : Acrolein has fruity odour and can be used to test glycerol's presence.

Choose the correct option.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q12. Numerical

Open solution

Among the following species

$N_2,\ N_2^+, N_2^-, N_2^{2-}, O_2, O_2^+, O_2^-, O_2^{2-}$

the number of species showing diamagnetism is _____.

Q13. Numerical

Open solution

The enthalpy of combustion of propane, graphite and dihydrogen at $298\ K$ are $-2220.0\ kJ\ mol^{-1}$, $-393.5\ kJ\ mol^{-1}$ and $-285.8\ kJ\ mol^{-1}$ respectively. The magnitude of enthalpy of formation of propane (C_3H_8) is _____ $kJ\ mol^{-1}$. (Nearest integer)

Q14. Numerical

Open solution

The cell potential for $Zn\left|Zn^{2+}(aq)\right|\left|Sn^{x+}\right|Sn$ is $0.801\ V$ at $298\ K$. The reaction quotient for the above reaction is 10^{-2}. The number of electrons involved in the given electrochemical cell reaction is _____.

(Given : $E^o_{Zn^{2+}|Zn} = -0.763\ V, E^o_{Sn^{x+}|Sn} = +0.008\ V$ and $\frac{2.303RT}{F} = 0.06\ V$)

Q15. Numerical

Open solution

The half life for the decomposition of gaseous compound A is $240\ s$ when the gaseous pressure was 500 Torr initially. When the pressure was 250 Torr, the half life was found to be 4.0 min. The order of the reaction is _____. (Nearest integer)

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Q16. Numerical

Open solution

Consider the following metal complexes :

$[Co(NH_3)_6]^{3+}$

$[CoCl(NH_3)_5]^{2+}$

$[Co(CN)_6]^{3-}$

$[Co(NH_3)_5(H_2O)]^{3+}$

The spin-only magnetic moment value of the complex that absorbs light with shortest wavelength is _____ B. M. (Nearest integer)

Q17. Numerical

Open solution

Among Co^{3+}, Ti^{2+}, V^{2+} and Cr^{2+} ions, one if used as a reagent cannot liberate H_2 from dilute mineral acid solution, its spin-only magnetic moment in gaseous state is _____ B.M. (Nearest integer)

Q18. Numerical

Open solution

While estimating the nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from $0.25\ g$ of the compound neutralized $2.5\ mL$ of $2\ M\ H_2SO_4$. The percentage of nitrogen present in organic compound is _____.

Q19. Numerical

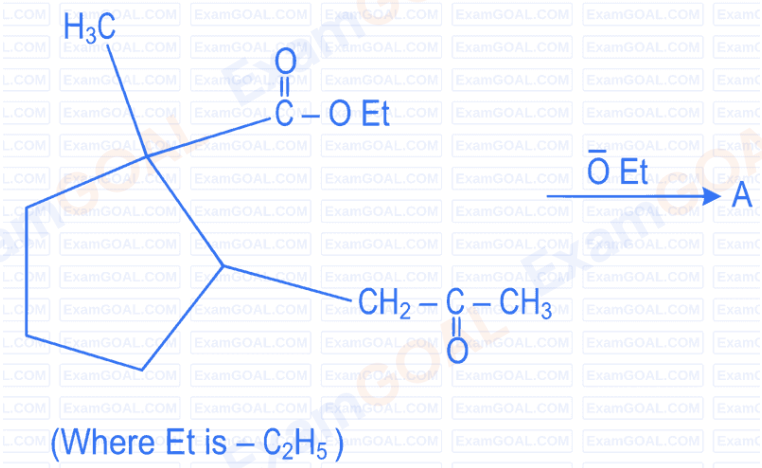
Open solution

The number of sp^3 hybridised carbons in an acyclic neutral compound with molecular formula C_4H_5N is _____.

Q20. Numerical

Open solution

In the given reaction



The number of chiral carbon/s in product A is _____.

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Mathematics 20 questions

Q21. MCQ (Single Correct Answer)

The total number of functions,

$f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$ such that $f(1) + f(2) = f(3)$, is equal to :

(A) 60 (B) 90 (C) 108 (D) 126

Open solution

Q22. MCQ (Single Correct Answer)

If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 + x^3 + x^2 + x + 1 = 0$, then $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021}$ is equal to :

(A) -4 (B) -1 (C) 1 (D) 4

Open solution

Q23. MCQ (Single Correct Answer)

For $n \in \mathbf{N}$, let $S_n = \left\{z \in \mathbf{C} : |z - 3 + 2i| = \frac{n}{4}\right\}$ and $T_n = \left\{z \in \mathbf{C} : |z - 2 + 3i| = \frac{1}{n}\right\}$. Then the number of elements in the set $\{n \in \mathbf{N} : S_n \cap T_n = \phi\}$ is :

(A) 0 (B) 2 (C) 3 (D) 4

Open solution

Q24. MCQ (Single Correct Answer)

The number of $\theta \in (0, 4\pi)$ for which the system of linear equations

$3(\sin 3\theta)x - y + z = 2$

$3(\cos 2\theta)x + 4y + 3z = 3$

$6x + 7y + 7z = 9$

has no solution, is :

(A) 6 (B) 7 (C) 8 (D) 9

Open solution

Q25. MCQ (Single Correct Answer)

If $\lim\limits_{n\to\infty}\left(\sqrt{n^2 - n - 1} + n\alpha + \beta\right) = 0$, then $8(\alpha + \beta)$ is equal to :

(A) 4 (B) -8 (C) -4 (D) 8

Open solution

Q26. MCQ (Single Correct Answer)

If the absolute maximum value of the function $f(x) = \left(x^2 - 2x + 7\right)e^{\left(4x^3 - 12x^2 - 180x + 31\right)}$ in the interval $[-3, 0]$ is $f(\alpha)$, then :

(A) $\alpha = 0$ (B) $\alpha = -3$ (C) $\alpha \in (-1, 0)$ (D) $\alpha \in (-3, -1]$

Open solution

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Q27. MCQ (Single Correct Answer)

The curve $y(x) = ax^3 + bx^2 + cx + 5$ touches the x-axis at the point $\mathrm{P}(-2, 0)$ and cuts the y-axis at the point Q, where y' is equal to 3 . Then the local maximum value of $y(x)$ is:

(A) $\frac{27}{4}$ (B) $\frac{29}{4}$ (C) $\frac{37}{4}$ (D) $\frac{9}{2}$

Open solution

Q28. MCQ (Single Correct Answer)

The area of the region given by

$A = \{(x, y) : x^2 \le y \le \min\{x + 2, 4 - 3x\}\}$ is :

(A) $\frac{31}{8}$ (B) $\frac{17}{6}$ (C) $\frac{19}{6}$ (D) $\frac{27}{8}$

Open solution

Q29. MCQ (Single Correct Answer)

For any real number x, let $[x]$ denote the largest integer less than equal to x. Let f be a real valued function defined on the interval $[-10, 10]$ by $f(x) = \begin{cases} x - [x], & \text{if } [x] \text{ is odd} \\ 1 + [x] - x, & \text{if } [x] \text{ is even} \end{cases}$. Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x \, dx$ is :

(A) 4 (B) 2 (C) 1 (D) 0

Open solution

Q30. MCQ (Single Correct Answer)

The slope of the tangent to a curve $C : y = y(x)$ at any point (x, y) on it is $\frac{2e^{2x} - 6e^{-x} + 9}{2 + 9e^{-2x}}$. If C passes through the points $\left(0, \frac{1}{2} + \frac{\pi}{2\sqrt{2}}\right)$ and $\left(\alpha, \frac{1}{2}e^{2\alpha}\right)$, then e^{α} is equal to :

(A) $\frac{3+\sqrt{2}}{3-\sqrt{2}}$ (B) $\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

(C) $\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$ (D) $\frac{\sqrt{2}+1}{\sqrt{2}-1}$

Open solution

Q31. MCQ (Single Correct Answer)

The general solution of the differential equation $\left(x - y^2\right)\mathrm{d}x + y\left(5x + y^2\right)\mathrm{d}y = 0$ is :

(A) $\left(y^2 + x\right)^4 = \mathrm{C}\left|\left(y^2 + 2x\right)^3\right|$ (B) $\left(y^2 + 2x\right)^4 = C\left|\left(y^2 + x\right)^3\right|$

(C) $\left|\left(y^2 + x\right)^3\right| = \mathrm{C}\left(2y^2 + x\right)^4$ (D) $\left|\left(y^2 + 2x\right)^3\right| = C\left(2y^2 + x\right)^4$

Open solution

Q32. MCQ (Single Correct Answer)

A line, with the slope greater than one, passes through the point $A(4, 3)$ and intersects the line $x - y - 2 = 0$ at the point B. If the length of the line segment AB is $\frac{\sqrt{29}}{3}$, then B also lies on the line :

(A) $2x + y = 9$ (B) $3x - 2y = 7$ (C) $x + 2y = 6$ (D) $2x - 3y = 3$

Open solution

Q33. MCQ (Single Correct Answer)

Let the locus of the centre $(\alpha, \beta), \beta > 0$, of the circle which touches the circle $x^2 + (y - 1)^2 = 1$ externally and also touches the x-axis be L. Then the area bounded by L and the line $y = 4$ is:

(A) $\frac{32\sqrt{2}}{3}$ (B) $\frac{40\sqrt{2}}{3}$ (C) $\frac{64}{3}$ (D) $\frac{32}{3}$

Open solution

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Q34. MCQ (Single Correct Answer)

Open solution

Let ABC be a triangle such that $\overrightarrow{BC}=\vec{a}, \overrightarrow{CA}=\vec{b}, \overrightarrow{AB}=\vec{c}, |\vec{a}|=6\sqrt{2}, |\vec{b}|=2\sqrt{3}$ and $\vec{b}\cdot\vec{c}=12$. Consider the statements :

$(S1): |(\vec{a}\times\vec{b})+(\vec{c}\times\vec{b})|-|\vec{c}|=6(2\sqrt{2}-1)$

$(S2): \angle ACB=\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Then

(A) both (S1) and (S2) are true

(B) only (S1) is true

(C) only (S2) is true

(D) both (S1) and (S2) are false

Q35. MCQ (Single Correct Answer)

Open solution

If the numbers appeared on the two throws of a fair six faced die are α and β, then the probability that $x^2+\alpha x+\beta>0$, for all $x\in\mathbf{R}$, is :

(A) $\frac{17}{36}$ (B) $\frac{4}{9}$ (C) $\frac{1}{2}$ (D) $\frac{19}{36}$

Q36. Numerical

Open solution

Let $A=\begin{pmatrix}2 & -1 & -1\\ 1 & 0 & -1\\ 1 & -1 & 0\end{pmatrix}$ and $B=A-I$. If $\omega=\frac{\sqrt{3}i-1}{2}$, then the number of elements in the set $\{n\in\{1,2,\ldots,100\}: A^n+(\omega B)^n=A+B\}$ is equal to _____.

Q37. Numerical

Open solution

The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is _____.

Q38. Numerical

Open solution

If the maximum value of the term independent of t in the expansion of $\left(t^2x^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}, x\geqslant 0$, is K, then $8\,K$ is equal to _____.

Q39. Numerical

Open solution

Let a, b be two non-zero real numbers. If p and r are the roots of the equation $x^2-8ax+2a=0$ and q and s are the roots of the equation $x^2+12\,bx+6\,b=0$, such that $\frac{1}{p}, \frac{1}{q}, \frac{1}{r}, \frac{1}{s}$ are in A.P., then $a^{-1}-b^{-1}$ is equal to _____.

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Q40. Numerical

Let $f(x)=\begin{cases}|4x^2-8x+5|, & \text{if } 8x^2-6x+1 \geqslant 0 \\ [4x^2-8x+5], & \text{if } 8x^2-6x+1<0,\end{cases}$ where $[\alpha]$ denotes the greatest integer less than or equal to α. Then the number of points in $\mathbf{R}$ where f is not differentiable is _____.

Open solution

Physics 28 questions

Q41. MCQ (Single Correct Answer)

If momentum [P], area $[\mathrm{A}]$ and time $[\mathrm{T}]$ are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is :

(A) $\left[\mathrm{P\,A^{-1}\,T^{0}}\right]$

(B) $\left[\mathrm{P\,A\,T^{-1}}\right]$

(C) $\left[\mathrm{P\,A^{-1}\,T}\right]$

(D) $\left[\mathrm{P\,A^{-1}\,T^{-1}}\right]$

Open solution

Q42. MCQ (Single Correct Answer)

Which of the following physical quantities have the same dimensions?

(A) Electric displacement $(\vec{\mathrm{D}})$ and surface charge density

(B) Displacement current and electric field

(C) Current density and surface charge density

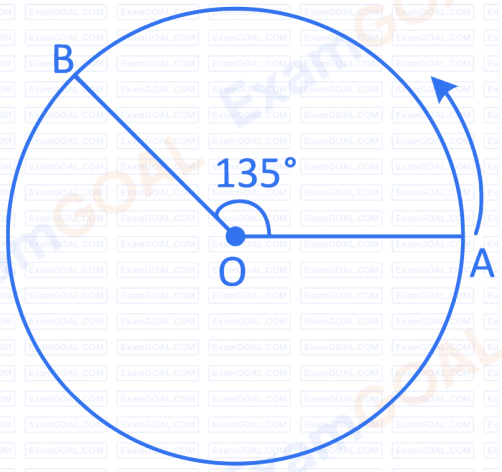
(D) Electric potential and energy

Open solution

Q43. MCQ (Single Correct Answer)

A person moved from A to B on a circular path as shown in figure. If the distance travelled by him is $60\ \mathrm{m}$, then the magnitude of displacement would be :

(Given $\cos 135^{\circ}=-0.7$)



(A) 42 m

(B) 47 m

(C) 19 m

(D) 40 m

Open solution

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Q44. MCQ (Single Correct Answer)

A body of mass $0.5\ \mathrm{kg}$ travels on straight line path with velocity $v = (3x^2 + 4)\mathrm{m/s}$. The net workdone by the force during its displacement from $x = 0$ to $x = 2\ \mathrm{m}$ is :

(A) 64 J (B) 60 J (C) 120 J (D) 128 J

Open solution

Q45. MCQ (Single Correct Answer)

A solid cylinder and a solid sphere, having same mass M and radius R, roll down the same inclined plane from top without slipping. They start from rest. The ratio of velocity of the solid cylinder to that of the solid sphere, with which they reach the ground, will be :

(A) $\sqrt{\frac{5}{3}}$ (B) $\sqrt{\frac{4}{5}}$ (C) $\sqrt{\frac{3}{5}}$ (D) $\sqrt{\frac{14}{15}}$

Open solution

Q46. MCQ (Single Correct Answer)

Three identical particles A, B and C of mass $100\ \mathrm{kg}$ each are placed in a straight line with $\mathrm{AB} = \mathrm{BC} = 13\ \mathrm{m}$. The gravitational force on a fourth particle P of the same mass is F, when placed at a distance $13\ \mathrm{m}$ from the particle B on the perpendicular bisector of the line AC. The value of F will be approximately :

(A) 21 G (B) 100 G (C) 59 G (D) 42 G

Open solution

Q47. MCQ (Single Correct Answer)

A certain amount of gas of volume V at $27^{\circ}\mathrm{C}$ temperature and pressure $2 \times 10^7 \mathrm{Nm}^{-2}$ expands isothermally until its volume gets doubled. Later it expands adiabatically until its volume gets redoubled. The final pressure of the gas will be (Use $\gamma = 1.5$) :

(A) $3.536 \times 10^5\ \mathrm{Pa}$ (B) $3.536 \times 10^6\ \mathrm{Pa}$ (C) $1.25 \times 10^6\ \mathrm{Pa}$ (D) $1.25 \times 10^5\ \mathrm{Pa}$

Open solution

Q48. MCQ (Single Correct Answer)

Following statements are given :

(A) The average kinetic energy of a gas molecule decreases when the temperature is reduced.

(B) The average kinetic energy of a gas molecule increases with increase in pressure at constant temperature.

(C) The average kinetic energy of a gas molecule decreases with increase in volume.

(D) Pressure of a gas increases with increase in temperature at constant pressure.

(E) The volume of gas decreases with increase in temperature.

Choose the correct answer from the options given below :

(A) (A) and (D) only (B) (A), (B) and (D) only

(C) (B) and (D) only (D) (A), (B) and (E) only

Open solution

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Q49. MCQ (Single Correct Answer)

Open solution



In figure (A), mass '$2\,\mathrm{m}$' is fixed on mass 'm' which is attached to two springs of spring constant k. In figure (B), mass 'm' is attached to two springs of spring constant 'k' and '$2\mathrm{k}$'. If mass 'm' in (A) and in (B) are displaced by distance 'x' horizontally and then released, then time period T_1 and T_2 corresponding to (A) and (B) respectively follow the relation.

(A) $\frac{\mathrm{T}_1}{\mathrm{T}_2}=\frac{3}{\sqrt{2}}$

(B) $\frac{\mathrm{T}_1}{\mathrm{T}_2}=\sqrt{\frac{3}{2}}$

(C) $\frac{\mathrm{T}_1}{\mathrm{T}_2}=\sqrt{\frac{2}{3}}$

(D) $\frac{T_1}{T_2}=\frac{\sqrt{2}}{3}$

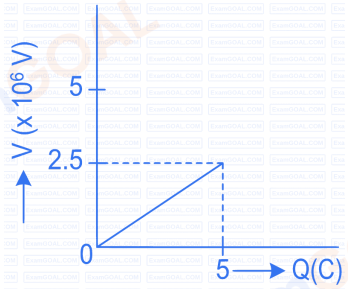
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Q50. MCQ (Single Correct Answer)

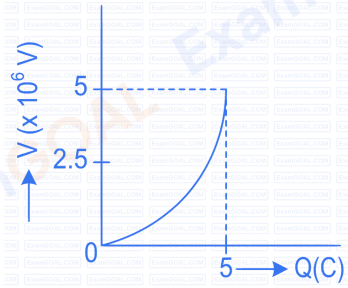
Open solution

A condenser of $2\ \mu\mathrm{F}$ capacitance is charged steadily from 0 to $5\ \mathrm{C}$. Which of the following graph represents correctly the variation of potential difference (V) across it's plates with respect to the charge (Q) on the condenser?

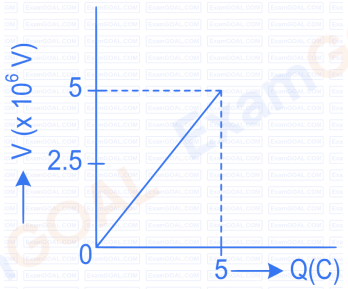
(A)



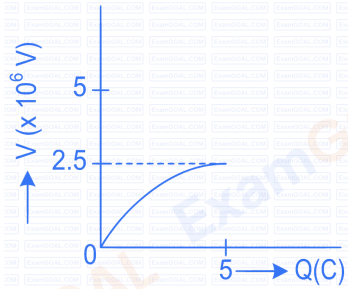
(B)



(C)



(D)



Q51. MCQ (Single Correct Answer)

Open solution

Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is $6:5$ and their respective masses ratio is $9:4$. Then, the ratio of their charges will be :

(A) 8 : 5 (B) 5 : 4 (C) 5 : 3 (D) 8 : 7

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Q52. MCQ (Single Correct Answer)

To increase the resonant frequency in series LCR circuit,

(A) source frequency should be increased.

(B) another resistance should be added in series with the first resistance.

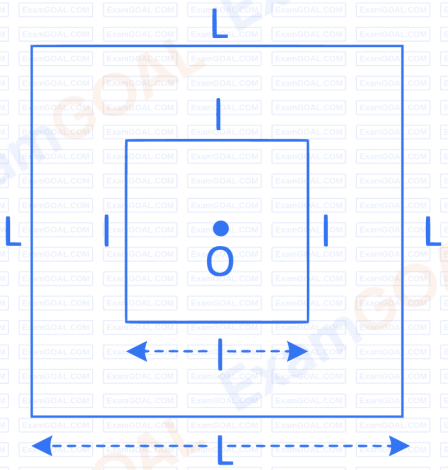
(C) another capacitor should be added in series with the first capacitor.

(D) the source frequency should be decreased.

Open solution

Q53. MCQ (Single Correct Answer)

A small square loop of wire of side l is placed inside a large square loop of wire $\mathrm{L}(\mathrm{L} >> l)$. Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is :



(A) $\frac{2\sqrt{2}\mu_0 \mathrm{L}^2}{\pi l}$

(B) $\frac{\mu_0 l^2}{2\sqrt{2}\pi \mathrm{L}}$

(C) $\frac{2\sqrt{2}\mu_0 l^2}{\pi \mathrm{L}}$

(D) $\frac{\mu_0 \mathrm{L}^2}{2\sqrt{2}\pi l}$

Open solution

Q54. MCQ (Single Correct Answer)

The rms value of conduction current in a parallel plate capacitor is $6.9\ \mu\mathrm{A}$. The capacity of this capacitor, if it is connected to $230\ \mathrm{V}$ ac supply with an angular frequency of $600\ \mathrm{rad/s}$, will be :

(A) 5 pF (B) 50 pF (C) 100 pF (D) 200 pF

Open solution

Q55. MCQ (Single Correct Answer)

Which of the following statement is correct?

(A) In primary rainbow, observer sees red colour on the top and violet on the bottom

(B) In primary rainbow, observer sees violet colour on the top and red on the bottom

(C) In primary rainbow, light wave suffers total internal reflection twice before coming out of water drops.

(D) Primary rainbow is less bright than secondary rainbow.

Open solution

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Q56. MCQ (Single Correct Answer)

Time taken by light to travel in two different materials A and B of refractive indices μ_A and μ_B of same thickness is t_1 and t_2 respectively. If $t_2 - t_1 = 5 \times 10^{-10}$ s and the ratio of μ_A to μ_B is $1 : 2$. Then, the thickness of material, in meter is: (Given v_{A} and v_{B} are velocities of light in A and B materials respectively.)

Open solution

(A) $5 \times 10^{-10}\, v_{\mathrm{A}}\ \mathrm{m}$

(B) $5 \times 10^{-10}\ \mathrm{m}$

(C) $1.5 \times 10^{-10}\ \mathrm{m}$

(D) $5 \times 10^{-10}\, v_{\mathrm{B}}\ \mathrm{m}$

Q57. MCQ (Single Correct Answer)

A metal exposed to light of wavelength $800\ \mathrm{nm}$ and emits photoelectrons with a certain kinetic energy. The maximum kinetic energy of photo-electron doubles when light of wavelength $500\ \mathrm{nm}$ is used. The workfunction of the metal is : (Take hc $= 1230\ \mathrm{eV} - \mathrm{nm}$).

Open solution

(A) 1.537 eV

(B) 2.46 eV

(C) 0.615 eV

(D) 1.23 eV

Q58. MCQ (Single Correct Answer)

The momentum of an electron revolving in n^{th} orbit is given by :

(Symbols have their usual meanings)

Open solution

(A) $\frac{\mathrm{nh}}{2\pi \mathrm{r}}$

(B) $\frac{nh}{2r}$

(C) $\frac{\mathrm{nh}}{2\pi}$

(D) $\frac{2\pi r}{\mathrm{nh}}$

Q59. MCQ (Single Correct Answer)

The magnetic moment of an electron (e) revolving in an orbit around nucleus with an orbital angular momentum is given by :

Open solution

(A) $\vec{\mu}_{\mathrm{L}} = \frac{\overrightarrow{\mathrm{eL}}}{2\,\mathrm{m}}$

(B) $\vec{\mu}_{\mathrm{L}} = -\frac{\overrightarrow{\mathrm{eL}}}{2\,\mathrm{m}}$

(C) $\vec{\mu}_l = -\frac{\overrightarrow{eL}}{\mathrm{m}}$

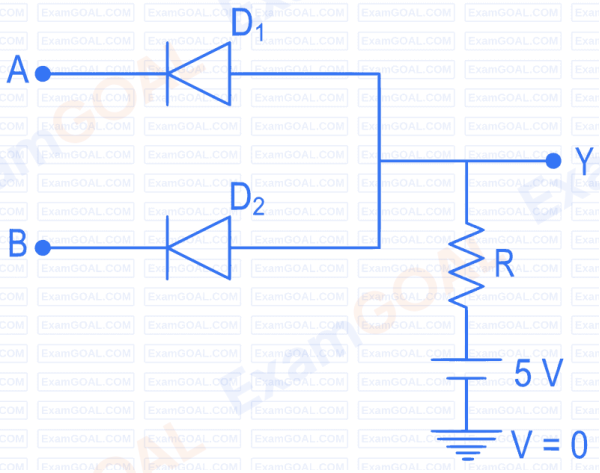
(D) $\vec{\mu}_l = \frac{\overrightarrow{2\mathrm{eL}}}{\mathrm{m}}$

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Q60. MCQ (Single Correct Answer)

Open solution

In the circuit, the logical value of $A = 1$ or $B = 1$ when potential at A or B is $5\ \text{V}$ and the logical value of $A = 0$ or $B = 0$ when potential at A or B is $0\ \text{V}$.



The truth table of the given circuit will be :

(A)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

(B)

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

(C)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

(D)

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

Q61. Numerical

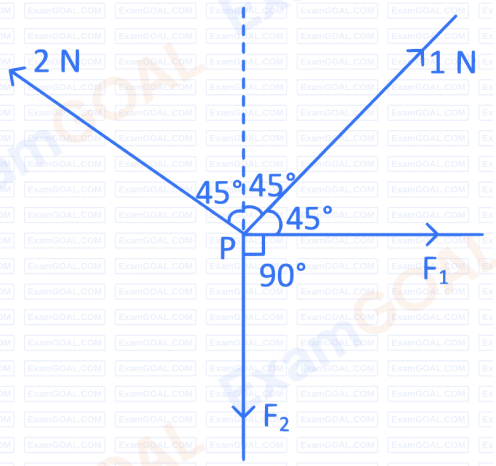
Open solution

A car is moving with speed of $150\ \text{km/h}$ and after applying the brake it will move $27\ \text{m}$ before it stops. If the same car is moving with a speed of one third the reported speed then it will stop after travelling _____ m distance.

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Q62. Numerical

Four forces are acting at a point P in equilibrium as shown in figure. The ratio of force F_1 to F_2 is $1 : x$ where $x =$ _____.



Open solution

Q63. Numerical

A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force F, its length increases by $5\ \mathrm{cm}$. Another wire of the same material of length $4\mathrm{L}$ and radius $4\mathrm{r}$ is pulled by a force $4\mathrm{F}$ under same conditions. The increase in length of this wire is _____ cm.

Open solution

Q64. Numerical

A unit scale is to be prepared whose length does not change with temperature and remains $20\ \mathrm{cm}$, using a bimetallic strip made of brass and iron each of different length. The length of both components would change in such a way that difference between their lengths remains constant. If length of brass is $40\ \mathrm{cm}$, then length of iron will be _____ cm. $\left(\alpha_{\text{iron}} = 1.2 \times 10^{-5}\ \mathrm{K}^{-1}\right.$ and $\left.\alpha_{\text{brass}} = 1.8 \times 10^{-5}\ \mathrm{K}^{-1}\right)$.

Open solution

Q65. Numerical

The volume charge density of a sphere of radius $6\ \mathrm{m}$ is $2\ \mu\mathrm{C\,cm}^{-3}$. The number of lines of force per unit surface area coming out from the surface of the sphere is _____ $\times 10^{10}\ \mathrm{NC}^{-1}$.

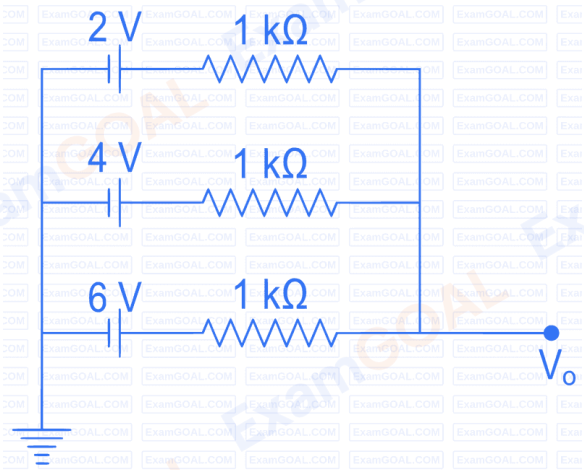
[Given : Permittivity of vacuum $\epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C}^2\ \mathrm{N}^{-1} - \mathrm{m}^{-2}$]

Open solution

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Q66. Numerical

In the given figure, the value of V_o will be _____ V.



Open solution

Q67. Numerical

Eight copper wire of length l and diameter d are joined in parallel to form a single composite conductor of resistance R. If a single copper wire of length $2l$ have the same resistance (R) then its diameter will be _____ d.

Open solution

Q68. Numerical

The energy band gap of semiconducting material to produce violet (wavelength = $4000\overset{o}{A}$) LED is _____ eV. (Round off to the nearest integer).

Open solution

Out of syllabus 22 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q69. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II

	List I		List II
(A)	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(I)	Cu
(B)	$CO(g) + 3H_2(g) \rightarrow CH_4(g) + H_2O(g)$	(II)	$Cu/ZnO - Cr_2O_3$
(C)	$CO(g) + H_2(g) \rightarrow HCHO(g)$	(III)	$Fe_xO_y + K_2O + Al_2O_3$
(D)	$CO(g) + 2H_2(g) \rightarrow CH_3OH(g)$	(IV)	Ni

Choose the correct answer from the options given below :

(A) $(A) - (II), (B) - (IV), (C) - (I), (D) - (III)$

(B) $(A) - (II), (B) - (I), (C) - (IV), (D) - (III)$

(C) $(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$

(D) $(A) - (III), (B) - (I), (C) - (IV), (D) - (II)$

Q70. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The compound(s) that is (are) removed as slag during the extraction of copper is :

(A) CaO

(B) FeO

(C) Al_2O_3

(D) ZnO

(E) NiO

Choose the correct answer from the options given below:

(A) (C), (D) only

(B) (A), (B), (E) only

(C) (A), (B) only

(D) (B) only

Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The reaction of H_2O_2 with potassium permanganate in acidic medium leads to the formation of mainly :

(A) Mn^{2+} **(B)** Mn^{4+} **(C)** Mn^{3+} **(D)** Mn^{6+}

Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Choose the correct order of density of the alkali metals :

(A) Li < K < Na < Rb < Cs

(B) Li < Na < K < Rb < Cs

(C) Cs < Rb < K < Na < Li

(D) Li < Na < K < Cs < Rb

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Q73. MCQ (Single Correct Answer) · **Chemistry**

The geometry around boron in the product 'B' formed from the following reaction is

$$BF_3 + NaH \xrightarrow{450\,K} A + NaF$$

$$A + NMe_3 \rightarrow B$$

(A) trigonal planar (B) tetrahedral (C) pyramidal (D) square planar

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

The interhalogen compound formed from the reaction of bromine with excess of fluorine is a :

(A) hypohalite (B) halate (C) perhalate (D) halite

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

The photochemical smog does not generally contain :

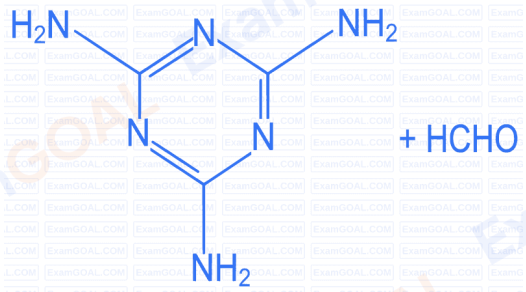
(A) NO (B) NO_2 (C) SO_2 (D) HCHO

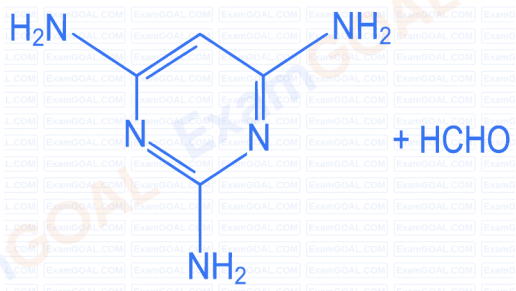
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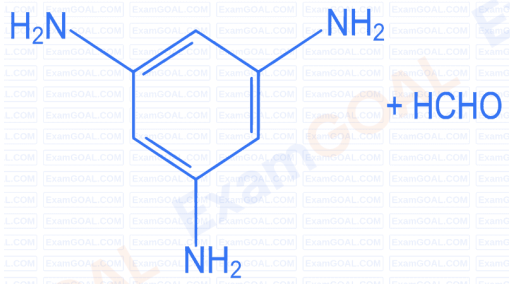
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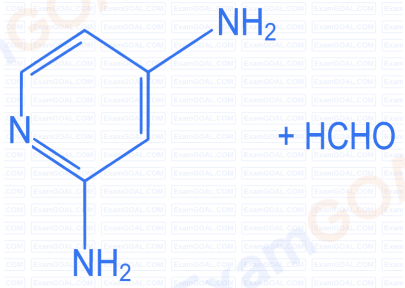
Q76. MCQ (Single Correct Answer) · **Chemistry**

Melamine polymer is formed by the condensation of:

(A) 

(B) 

(C) 

(D) 

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Drugs used to bind to receptors, inhibiting is natural function and blocking a message are called:

(A) Agonists

(B) Antagonists

(C) Allosterists

(D) Anti histaminists

Open solution

Q78. Numerical · **Chemistry**

The pressure of a moist gas at $27^{\circ}\mathrm{C}$ is $4\ \mathrm{atm}$. The volume of the container is doubled at the same temperature. The new pressure of the moist gas is _____ $\times 10^{-1}\ \mathrm{atm}$. (Nearest integer)

(Given : The vapour pressure of water at $27^{\circ}\mathrm{C}$ is $0.4\ \mathrm{atm}$.)

Open solution

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Q79. MCQ (Single Correct Answer) · **Mathematics**

Let P be the plane containing the straight line $\frac{x-3}{9}=\frac{y+4}{-1}=\frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$. If d is the distance of P from the point $(2,-5,11)$, then d^2 is equal to :

(A) $\frac{147}{2}$ (B) 96 (C) $\frac{32}{3}$ (D) 54

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

If the sum and the product of mean and variance of a binomial distribution are 24 and 128 respectively, then the probability of one or two successes is :

(A) $\frac{33}{2^{32}}$ (B) $\frac{33}{2^{29}}$ (C) $\frac{33}{2^{28}}$ (D) $\frac{33}{2^{27}}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

The number of solutions of $|\cos x|=\sin x$, such that $-4\pi \leq x \leq 4\pi$ is :

(A) 4 (B) 6 (C) 8 (D) 12

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower in two parts such that $QR=15\ \mathrm{m}$. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is :

(A) $5(2\sqrt{3}+3)\ \mathrm{m}$ (B) $5(\sqrt{3}+3)\ \mathrm{m}$ (C) $10(\sqrt{3}+1)\ \mathrm{m}$ (D) $10(2\sqrt{3}+1)\ \mathrm{m}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Which of the following statements is a tautology ?

(A) $((\sim \mathrm{p}) \vee \mathrm{q}) \Rightarrow \mathrm{p}$ (B) $p \Rightarrow ((\sim p) \vee q)$

(C) $((\sim p) \vee q) \Rightarrow q$ (D) $q \Rightarrow ((\sim p) \vee q)$

Open solution

Q84. Numerical · **Mathematics**

Let $a_1=b_1=1, a_n=a_{n-1}+2$ and $b_n=a_n+b_{n-1}$ for every natural number $n \geqslant 2$. Then $\sum_{n=1}^{15} a_n . b_n$ is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

If $\lim_{n\to\infty} \frac{(n+1)^{k-1}}{n^{k+1}}[(nk+1)+(nk+2)+\ldots+(nk+n)]$ $=33 \cdot \lim_{n\to\infty} \frac{1}{n^{k+1}} \cdot \left[1^k+2^k+3^k+\ldots+n^k\right]$, then the integral value of k is equal to _____

Open solution

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Q86. Numerical · **Mathematics**

Let the equation of two diameters of a circle $x^2 + y^2 - 2x + 2fy + 1 = 0$ be $2px - y = 1$ and $2x + py = 4p$. Then the slope m $\in (0, \infty)$ of the tangent to the hyperbola $3x^2 - y^2 = 3$ passing through the centre of the circle is equal to _____.

Open solution

Q87. Numerical · **Mathematics**

The sum of diameters of the circles that touch (i) the parabola $75x^2 = 64(5y - 3)$ at the point $\left(\frac{8}{5}, \frac{6}{5}\right)$ and (ii) the y-axis, is equal to _____.

Open solution

Q88. Numerical · **Mathematics**

The line of shortest distance between the lines $\frac{x-2}{0} = \frac{y-1}{1} = \frac{z}{1}$ and $\frac{x-3}{2} = \frac{y-5}{2} = \frac{z-1}{1}$ makes an angle of $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ with the plane $\mathrm{P}: \mathrm{a}x - y - z = 0$, $(a > 0)$. If the image of the point $(1, 1, -5)$ in the plane P is (α, β, γ), then $\alpha + \beta - \gamma$ is equal to _____.

Open solution

Q89. Numerical · **Physics**

An observer is riding on a bicycle and moving towards a hill at $18\,\mathrm{kmh}^{-1}$. He hears a sound from a source at some distance behind him directly as well as after its reflection from the hill. If the original frequency of the sound as emitted by source is $640\,\mathrm{Hz}$ and velocity of the sound in air is $320\,\mathrm{m/s}$, the beat frequency between the two sounds heard by observer will be _____ Hz.

Open solution

Q90. Numerical · **Physics**

The required height of a TV tower which can cover the population of 6.03 lakh is h. If the average population density is 100 per square km and the radius of earth is $6400\,\mathrm{km}$, then the value of h will be _____ m.

Open solution

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Answer Key

Chemistry

1. **C**	2. **C**	3. **C**	4. **C**	5. **D**
6. **D**	7. **B**	8. **D**	9. **C**	10. **A**
11. **C**	12. **2**	13. **104**	14. **4**	15. **1**
16. **0**	17. **5**	18. **56**	19. **1**	20. **2**

Mathematics

21. **B**	22. **B**	23. **D**	24. **B**	25. **C**
26. **B**	27. **A**	28. **B**	29. **A**	30. **B**
31. **A**	32. **C**	33. **C**	34. **C**	35. **A**
36. **17**	37. **1492**	38. **6006**	39. **38**	40. **3**

Physics

41. **A**	42. **A**	43. **B**	44. **B**	45. **D**
46. **B**	47. **B**	48. *Scan QR*	49. **A**	50. **A**
51. **B**	52. **C**	53. **C**	54. **B**	55. **A**
56. **A**	57. **C**	58. **A**	59. **B**	60. **A**
61. **3**	62. **3**	63. **5**	64. **60**	65. **45**
66. **4**	67. **4**	68. **3**		

Out of syllabus

69. **C**	70. **D**	71. **A**	72. **A**	73. **B**
74. **B**	75. **C**	76. **A**	77. **B**	78. **22**
79. **C**	80. **C**	81. **C**	82. **A**	83. **D**
84. **27560**	85. **5**	86. **2**	87. **10**	88. **3**
89. **20**	90. **150**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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